



OVERVIEW

AMS Derive | A structured, multi-stage evaluation system for identifying first-principles thinkers in quantitative and algorithmic domains.

AMS Derive is a rigorous signal-generation system designed to identify elite mathematical minds. It is built to surface individuals who can reason from first principles, model uncertainty, and structure elegant solutions to complex open-ended problems.

It tests how you think. Not what you have memorised. Not how fast you can pattern-match. The evaluation is designed to reward structured reasoning, probabilistic thinking, and the ability to derive, not recall.

Three rounds. Progressively open-ended. Final round in-person at offline finals. Built for the top end of engineering and mathematics talent across institutions.

COMPETITION STRUCTURE

Three rounds. Each stage tests a distinct layer of thinking ability, from mathematical precision to open-ended system reasoning. Progression is not guaranteed. Selection at each stage is based on signal quality, not volume.

Round 1 | PRIOR | Mathematical Modeling · Online · 3 Hours

Problems are drawn from probability, stochastic processes, expected value, and distribution theory. No templates are provided. Participants are expected to build models from first principles and show their reasoning in full.

Evaluated on: Correctness · Clarity of reasoning · Rigour of derivation

Round 2 | POSTERIOR | Algorithmic + Quant Thinking · Online · 3 Hours

Competitive-programming style problems with a quantitative layer. Correct answers are not enough, solutions must be efficient, well-structured, and handle edge cases under constraints. Problems combine algorithmic thinking with probabilistic reasoning.

Evaluated on: Efficiency · Correctness under constraints · Approach clarity

Round 3 | CONVERGENCE | Final Round · In-Person, Offline Finals · Top X Participants

Selected participants are invited to the offline finals. Teams are formed on-site. Problems are open-ended, real-world, and deliberately ambiguous, co-designed with partner firms. There is no single correct answer. Evaluation is based on how the problem is structured, reasoned through, and communicated.

Problem type: Real-world systems · Market structures · Order book mechanics

Evaluated on: Problem framing · Reasoning depth · Execution · Communication

TIMELINE

All dates are tentative and subject to confirmation. Follow amsderive.in for live updates.



Registered participants will be notified directly via email at each stage transition.

ELIGIBILITY

AMS Derive is built exclusively for university students demonstrating strong first-principles reasoning. While no strict coding or financial background is explicitly required, individuals must possess deep mathematical maturity and logical structuring ability to clear the initial assessments.

Demographics | Open to Current Undergraduate & Postgraduate Students

Participation is extended to students currently enrolled in B.Tech, M.Tech, B.S., M.S., Ph.D., or equivalent technical degree programs across **India**. There are no restrictions based on college tier, branch, or current year of study. Evaluation relies entirely on signal generation.

Format | Individual Qualification · Team-Based Convergence

Round 1 (Prior) and Round 2 (Posterior) are strictly individual. Collaboration, discussion, or the use of generative AI or external templates is prohibited and actively monitored. Participants who qualify for Round 3 (Convergence) will be automatically clustered into dynamic teams on-site at the offline finals to collaboratively model market structure problems.

Restrictions | No Full-Time Quants/Traders · 1-Year Grace Period for Grads

This circuit is primarily designed for students (**actively enrolled for Bachelor’s, Master’s, PHD Programs or in High School**). However, recent alumni (graduated within the last 12 months) are eligible to participate. Individuals who graduated 2 or more years ago are strictly disqualified.

EVALUATION PHILOSOPHY

Most evaluation systems reward speed and recall. AMS Derive does not. The design assumption here is that the ability to retrieve a known method is a weak signal. The ability to construct a solution from underlying principles, under constraint, without a template, is the signal worth measuring.

Principle 1 | Reasoning Over Recall

Problems are designed so that memorised solutions do not transfer directly. Participants are expected to reason through unfamiliar configurations, not apply known patterns. Partial derivations with sound logic are valued over complete answers with no visible reasoning.

Principle 2 | Modeling Ability Over Speed

Time pressure exists, but it is not the primary filter. The evaluation prioritises the quality of a participant's model, how well they decompose a problem, identify the relevant variables, and structure their approach. A clean, well-reasoned partial solution outranks a fast but shallow complete one.

Principle 3 | Structured Thinking Under Uncertainty

Especially in Rounds 2 and 3, problems are deliberately underspecified. Participants must identify what is unknown, make explicit assumptions, and reason forward. This mirrors how problems are actually encountered in quantitative and research environments, ambiguous inputs, no guarantee of a closed-form answer.

What is rewarded: Logical decomposition · Explicit assumptions · Derivation clarity · Structured communication

What is not rewarded: Pattern matching · Speed without rigour · Memorised templates

SYLLABUS

The following topics form the working scope of AMS Derive. Problems are not restricted to this list, but participants who have internalised these areas at a reasoning level, not just a computational one, will be well-positioned across all three rounds.

Mathematics | Probability, Stochastic Processes & Analytical Foundations

The mathematical core of AMS Derive. Problems in Round 1 draw heavily from this area. Comfort with derivation, not just formula application, is assumed.

Probability: Sample spaces · Conditional probability · Bayes' theorem · Independence

Random Variables: Discrete and continuous distributions · PMF, PDF, CDF · Joint distributions

Expectation & Variance: Linearity of expectation · Moments · MGFs · Tail bounds

Markov Chains: Transition matrices · Stationary distributions · Absorption and recurrence

Stochastic Processes: Random walks · Martingales · Basic Brownian motion intuition

Algorithms | Combinatorial Reasoning & Computational Problem Solving

Round 2 problems sit at the intersection of algorithmic thinking and quantitative reasoning. Standard competitive programming knowledge is necessary but not sufficient, problems are designed to require probabilistic layers on top of classical structures.

Graphs: BFS, DFS · Shortest paths · Trees · Flows and matchings

Dynamic Programming: Optimal substructure · State design · DP on sequences, trees, graphs

Greedy Algorithms: Exchange arguments · Interval scheduling · Correctness proofs

Optimization: Constraint handling · Combinatorial search · Expected-cost minimization

Quant Concepts | Applied Intuition for Real-World Systems

These concepts are not prerequisites for Rounds 1 or 2. They become relevant in Round 3, where problems are grounded in real market and system design contexts. Deep domain knowledge is not required, structured thinking applied to these domains is.

Order Books: Bid-ask structure · Limit and market orders · Price impact intuition

Pricing Intuition: No-arbitrage reasoning · Risk-neutral thinking · Relative valuation

Arbitrage Logic: Identifying mispricing · Constructing hedges · Reasoning about market efficiency

RULES & GUIDELINES

These rules are non-negotiable. Violations at any stage result in immediate disqualification. AMS reserves the right to verify, investigate, and nullify results at its discretion.

OBS Recording | Mandatory for All Online Rounds, No Exceptions

All participants must actively record their screen using OBS Studio (or equivalent) for the entire duration of the online assessments. For Round 1, only a continuous screen capture is required. From Round 2 onwards, both screen capture and a continuous webcam feed are strictly mandatory. Recording must begin before the problem set is accessed and run continuously until submission. Any submission received without a valid recording will be treated as void.

Round 1 Required: OBS Studio (or equivalent) · Continuous screen capture

Round 2 Required: OBS Studio · Continuous screen capture · Continuous webcam feed

Submission: Recording file must be uploaded with the solution before the deadline

Technical Verification | Top Candidates Selected After Each Online Round

Following Round 1 and Round 2, the top X candidates from each round will undergo a mandatory technical verification session conducted remotely by the AMS team. Participants will be asked to walk through their submitted solutions, explain their reasoning, and respond to follow-up questions. Solutions that cannot be defended will be disqualified. Advancement to the next round is contingent on passing verification.

Format: Short remote session · Solution walkthrough · Reasoning defence

Applies to: Top X candidates post Round 1 · Top X candidates post Round 2

Outcome: Failure to defend = disqualification, irrespective of score

Integrity | Individual Rounds Are Strictly Solo

Rounds 1 and 2 are individual assessments. Discussion with other participants, use of generative AI tools, pre-written solution templates, or external assistance of any kind is strictly prohibited. Identical or structurally similar submissions will be flagged and investigated. Both parties in a plagiarism case are disqualified.

Submissions | Format, Deadline, and Completeness

Submissions must follow the specified format for each round. Late submissions will not be accepted under any circumstances. Incomplete submissions, missing reasoning, missing recording, or missing required files, will not be evaluated. It is the participant's responsibility to verify their submission before the deadline.

Disqualification conditions: Plagiarism · Missing OBS recording · AI-generated solutions · Late submission · Inability to defend solution in verification

FINAL ROUND DETAILS

The final round is a different kind of evaluation. It is not a longer version of the online rounds. It is in-person, team-based, and deliberately open-ended. The problems are real. The constraints are real. The expectation is that participants can think, structure, and communicate under pressure, not just solve.

Format | In-Person · Offline Finals · Top Participants Only

Finalists will be invited to attend the offline finals at a confirmed venue. Attendance is mandatory, participants who do not confirm within the stipulated window or fail to attend will forfeit their spot. Venue and logistical details will be communicated directly to confirmed finalists.

Selection: Top participants from Round 2 + verification · Invitation via email

Attendance: Confirmation required within specified deadline · No remote participation

Teams | Formed On-Site · No Pre-Registration of Teams

Teams are formed on the day of the final round. Participants do not choose their teams. The clustering is intentional, it tests the ability to collaborate with unfamiliar people, establish shared reasoning quickly, and produce a coherent output under time pressure. Pre-coordination with other finalists on team composition is not permitted.

Problem Structure | Open-Ended · Real-World · Co-Designed with Partner Firms

Problems are not drawn from textbooks. They are co-designed with partner firms and grounded in real-world system and market contexts. Problems are deliberately ambiguous, there is no single correct answer. Teams are evaluated on how they frame the problem, what assumptions they make explicit, how they model the system, and how clearly they communicate their conclusions.

Problem domains: Market structure · System design · Order flow reasoning · Uncertainty modeling

Judged on: Problem framing · Reasoning depth · Execution quality · Clarity of communication

What to Bring | Laptop · Charger · Nothing Else Required

Participants must bring a personal laptop with a working development or document environment. All problem materials will be provided on the day. No pre-prepared notes, templates, or resources are permitted. The final round tests thinking, not preparation of answers.

The offline finals are not just an evaluation. The format is designed to bring together a concentrated group of strong analytical minds , and that itself is valuable. Alongside the main event, finalists can expect structured chess and poker sessions, both of which are natural extensions of the probabilistic and strategic reasoning that AMS Derive selects for. There will also be dedicated time for networking with peers, AMS team members, and representatives from partner firms.

Side events: Chess · Poker (strategy-focused) · Open networking

Who you will meet: Fellow finalists · AMS team · Partner firm representatives

THE PINNACLE OF QUANTITATIVE REASONING

The **Top 3 Teams** emerge victorious and win the title of **AMS Derive Champions**.

[PRE-REGISTER AT AMSDERIVE.in](https://amsderive.in)